Network Services

Chapter 20: Sendmail

Chapter 21: NIS and LDAP

Chapter 22: NFS

Chapter 23: Samba

Chapter 20: Sendmail: Setting Up Mail Clients, Servers, and More

- Configuring sendmail on a client
- Configuring sendmail on a server
- How sendmail works
- Configuring sendmail
- SpamAssassin
- Webmail and mailing lists
- Setting up IMAP or POP3 server
- Authenticated relaying

Introduction to sendmail

- Using email requires three pieces of software
  - Client, called MUA (Mail User Agent)
    - mutt, KMail, Thunderbird, Outlook, Lotus Notes
    - Most can use POP, IMAP, or SMTP to receive, SMTP to send emails
  - Mail Transfer Agent (MTA) sends to destination server
    - sendmail
  - Mail Delivery Agent (MDA) puts email in recipient's mailbox
    - procmail
- Do not need to set up `sendmail` to send and receive email

What Can sendmail Do?

- Users usually maintain an account on an email server run by employer or ISP
  - Receive email using POP or IMAP protocols
- Can set up `sendmail` to
  - Receive email for an email server
  - Relay outbound mail to an SMTP server
  - Be an outbound server that sends directly to SMTP servers for domains receiving the email
- For email server, also set up POP or IMAP sever to interact with mail clients (MUAs)

Packages to Install

- sendmail (required)
- sendmail-cf (required to configure sendmail)
- squirrelmail (optional; provices webmail)
- spamassassin (optional; provides spam filtering)
- mailman (optional; provides mailing lists)
- dovecot (optional; provides IMAP and POP)
- Note: to configure SMTP server will need to open a firewall port (usually TCP port 25)

Configuring sendmail on a Client

- Test existing configuration by command
  - echo “test” | /usr/sbin/sendmail user@remote.host
- If doesn't work, configure sendmail to
  - Use remote SMTP server to relay outbound email
  - Only send email originating from local system
  - Not handle inbound email (use POP or IMAP)
- Edit /etc/mail/sendmail.mc and restart sendmail
  - Remove dnl and enter smtp server name in line:
  - `dnl define(`SMART_HOST', `smtp.your.provider')dnl`
  - Preserve back ticks (`) and quotes (')

Configuring sendmail on Server

- Configure sendmail to
  - Accept outbound email from local system only
  - Deliver outbound email directly to recipient's system, not using relay
  - Accepts inbound email from any system
- Edit following line in sendmail.mc removing Addr=127.0.0.1,
  - `DAEMON_OPTIONS(`Port=smtp,Addr=127.0.0.1, Name=MTA')dnl`
- Domain DNS also needs MX record pointing to server

How sendmail Works

- MUA passes email to sendmail
- `sendmail` separates email into headers/body and stores in `/var/spool/mqueue` directory
  - When email sent, files deleted
- Incoming mail stored in /var/spool/mail
  - Default `mbox` format
    - `maildir` format has each message in separate file
  - File per user named with user's username
  - Mail stored until collected by MUA
  - Log messages stored in /var/log/maillog

Aliases and Forwarding

- Three files can forward email: aliases, .forward, virtusertable
- `/etc/aliases` maps pseudousers to real users
  - `system:  root`
  - `abuse:  root, fred`
  - `admin:  sam, sally, aabdulla@bcit.ca`
  - `complaints: /dev/null`
- `~/.forward` forwards a user's email
  - One destination per line
  - Destination can be local user, email address, filename, or command preceded by pipe (|)

Configuring sendmail

- Sendmail configuration files are in `/etc/mail`
  - sendmail.cf is main configuration file
    - Edit `sendmail.mc`; macro processor `m4` generates sendmail.cf when `sendmail` restarted
    - Lines starting with `dnl` are comments (to end of line)
    - Strings are preceded with backtick (`` ` ``), closed with single quote (`'`)
  - Also `access`, `mailertable`, `virtusertable`
    - sendmail reads processed .db versions
- Full understanding of sendmail configuration requires *much* more than we cover
  - O'Reilly's sendmail 4th edition has 1308 pages!

Other Actions

- Masquerading – can change outbound messages so come from different user/domain
  - Intended for case where outbound server is in different domain than inbound server
  - Use MASQUERADE_AS, MASQUERATE_DOMAIN, and FEATURE(masquerade_entire_domain)
- Reduce spam by refusing mail from unknown hosts
  - Comment: FEATURE(`accept_unresolvable_domains')dnl
- Set up backup server
  - Uncomment: FEATURE(`relay_based_on_MX')
  - Set up additional MX records in DNS server

Other Files in /etc/mail

- `mailertable`: forward from one domain to another
  - Each line has domain name, white space, destination mailer
  - Mail sent to domain is forwarded to destination
- `access`: set up relay host
  - Specifies which systems can relay through this server
- `virtusertable`: serve email to multiple domains
  - Each line has address email was sent to, whitespace, and address to forward to
  - Address can be local user, email address, file, or pipe (|) followed by command

SpamAssasin

- Checks all inbound email before sent to users
- Individual users can also run from their mail clients
- Both use spamd daemon and spamc client
- SpamAssassin uses several techniques
  - Header analysis to make sure header is legitimate
  - Text analysis for characteristics of spam
  - Blacklists to see if sender is known for spam
  - Database (Vipul's Razor: razor.sourceforge.net)

Configuring SpamAssassin

- Make sure spamd is installed and running
- Edit /etc/mail/spamassassin/local.cf to configure globally
  - Users configure in ~/.spamassassin/user_prefs
- Change required_hits (default 5) if too many / too little emails are marked as spam
- Specify addresses never marked as spam using whitelist_from keyword
  - Addresses always marked as spam with blacklist_from keyword

Using SpamAssassin From Server

- Be sure you are using procmail as MDA
  - Default for Fedora
- Configure /etc/procmailrc (owned by root, 644)
  - DROPPRIVS=yes
  - :0 fw
  - | /usr/bin/spamc
- Inditates filter rule and procmail should wait for filter to complete and check exit code
- Filter sends email through spamc

Using SpamAssassin with Mail Client

- Need to set up two rules in mail client
  - Pass each piece of email to spamc
    - Spam not usually too large, perhaps filter based on size smaller than 200,000 bytes
    - SpamAssassin will add X-Spam-Flag with results
  - Filter email based on X-Spam-Flag
    - If spam, move to spam folder
- Exactly how to do this will depend on mail client
- Text shows example of KMail

WebMail

- Common to use web browser to read email
  - Gmail, HotMail, Yahoo! Mail
- SquirrelMail provides webmail services
  - Have to set up IMAP first and also Apache
  - Comes with plugins such as shared calendar
- To make SquirrelMail accessable from web create link
  - ln -s /usr/share/squirrelmail /var/www/html/mail
- Users get mail from http://*yourHost*/mail
- Use conf.pl script in /usr/share/squirrelmail/config to configure SquirrelMail
  - Set server's domain name at least

Mailing Lists

- Can be used to distribute mail to large group of people
  - Such as project team or special interest group
- Mailman provides mailing list support
  - Application in /usr/lib/mailman
  - Configuration in /etc/mailman/mm_cfg.py
    - Need to specify local domain in DEFAULT_UTL_HOST and DEFAULT_EMAIL_HOST
- Create new mailing list with newlist utility
  - /usr/lib/mailman/bin/newlist
- Will need to copy lines generated by newlist to end of /etc/aliases and run newaliases

Setting up IMAP or POP3 Server

- The dovecot package includes daemons that implement IMAP and POP protocols
  - Usually do not need to modify configuration in /etc/dovecot.conf
  - Installation script creates self-signed certificates that dovecot needs
  - Set up dovecot service to start automatically
    - Starts both IMAP and POP servers

Authenticated Relaying

- Can set up sendmail to authenticate before accepting email for processing
  - Can then email from any internet connection
  - sendmail can use secure connection
  - Outgoing mail server requires authentication so can not be used for spam
- Need to
  - Create SSL certificate
  - Enable SSL in sendmail
  - Tell email client to connect to SMTP using SSL

Using Self-Signed Certificate in sendmail

- As root, cd to /etc/pki/tls/certs
- Run make sendmail.pem and answer questions
- Enable SSL in sendmail by uncommenting lines in sendmail.mc
  - TRUST_AUTH_MECH...
  - define(`confAUTH_MECHANISMS', …
  - Lines defining confCACERT_PATH, confCACERT, confSERVER_CERT, confSERVER_KEY
  - DAEMON_OPTIONS(`Port=smtps, Name-TLSMTA, M=s')
- Enable SSL in Mail Client by choosing encryption SSL

Summary: sendmail

- sendmail daemon is an MTA (Mail Transfer Agent)
  - Can relay email to SMTP server or send directly to domain receiving the email
  - stores incoming messages in mail spool directory /var/spool/mail
- Configuration file is generated from sendmail.mc
- Administrator can set forwarding with /etc/aliases

Chapter 21: NIS and LDAP

- NIS (Network Information Service) simplifies maintenance of common administration files on a LAN
  - Can share files such as /etc/passwd
- LDAP (Lightweight Directory Access Protocol) can hold any read-mostly structured information requiring fast lookup
  - Names, addresses, lists of network services, authentication data
  - Windows uses LDAP for its Active Directory

How NIS Works

- NIS uses NIS domain name to characterize set of resources
  - Different than DNS domain name!
- Each NIS domain has master server and may have slave servers
- Information distributed can come from /etc
  - group, gshadow, hosts, passwd, printcap, rpc, services, shadow
- Information converted to map files to serve up

Setting up NIS Client

- Install yp-tools and ypbind
  - Configure ypbind to start in multiuser mode
- Specify NIS domain name in /etc/sysconfig/network file
  - NISDOMAIN=*nisdomainname*
- Edit /etc/yp.conf to specify NIS server
  - domain *nisdomain* server *server_name* or
  - ypserver *server_name*
- Restart ypbind

Testing NIS Setup

- Check domain with `ypwhich`
- Make sure rpc finds server
  - /usr/sbin/rpcinfo -u *server* ypserv
  - Should report server as ready and waiting
- Check ypbind has registered with portmap
  - /usr/sbin/rpcinfo -u localhost ypbind
  - Should report ready and waiting
- If problems, can run ypbind in forground
  - Stop ypbind service and run /sbin/ypbind -debug

Changing NIS Passwords

- If using NIS to distribute passwords
  - Use `yppasswd` to change user's password
  - passwd utility changes password in local files
    - OK for root account but not user accounts
  - If set up, user can also use yppasswd to change login shell and user information (GECOS)
    - `yppasswd -f` to change user information
- Add and remove users by making change on server
  - Run `ypinit` to update map

Setting Up NIS Server

- Install `ypserv` package and have it start up in multiuser mode
  - On master server only
    - have map server `ypxfrd` start when system enters multiuser mode
    - Have password update server `yppasswdd` start when system enters multiuser mode
- After configuring ypserv go start (in order)
  - Ypserv, ypxfrd, yppasswdd

NIS Server Setup Steps

- Specify NIS domain name in /etc/sysconfig/network file
  - NISDOMAIN=*nisdomainname*
- Edit /etc/ypserv.conf to configure NIS server
  - Rules have format *option: value*
- Create /var/yp/securenets for more security
- Edit /var/yp/Makefile to specify maps
- Start servers

/etc/ypserv.conf

- Options
  - files – maximum number of files cached
  - trusted_master – master a slave gets files from
  - xfer_check_port – YES requires master to run on privileged port (0-1023)
- Access rules – who can access which maps
  - format: *host:domain:map:security*
  - *host* is IP address (possibly with subnet mask)
  - *domain* is NIS domain
  - *map* is name of map rule applies to
  - *security* is none, port, or deny; port means privileged port

Create /var/yp/securenets to Enhance Security

- Prevents unauthorized systems from sending RPC requests and receiving NIS maps
  - Critical file to be protected is `shadow` file
  - Without `securenets` file, requests accepted from any system
- Each line has a netmask and IP address
  - NIS accepts request from any IP in these subnets
  - E.g. 255.255.255.0 142.232.18.0
  - E.g. 255.255.255.192 192.168.0.0
  - Must have localhost: 255.255.255.255 127.0.0.1

Edit /var/yp/Makefile to Specify Maps

- Makefile controls what maps are built
  - Run with ypinit command (don't run make)
  - `/usr/lib64/yp/ypinit -m` *or*
  - `/usr/lib64/yp/ypinit -s` *`master_server`*
- The all: target specifies which maps to build
  - all: passwd group hosts rpc services netid protocols mail \
  - # netgrp shadow publickey networks ethers bootparams printcap \
  - # amd.home auto.master auto.home auto.local passwd.adjunct \
  - # timezone locale netmasks
- First line is defaults, other lines are files for consideration

LDAP (Light-weight Directory Access Protocol)

- Designed to distribute information that changes infrequently (directory)
  - Fedora and RHEL use OpenLDAP
  - Optimized for fast lookup
- Data is organized into hierarchy
- LDAP clients can query and update information
- These slides give an overview of LDAP only

Directory Services

- Directories used to store information about users, accounts, and other objects
- Often have several incompatible directories
  - Mail required different account from windows account
  - Linux accounts separate from windows accounts
  - Web logins separate from OS logins
- Directory services try to combine information into one centrally managed database
  - All services and information about a network

Lightweight Directory Access Protocol (LDAP)

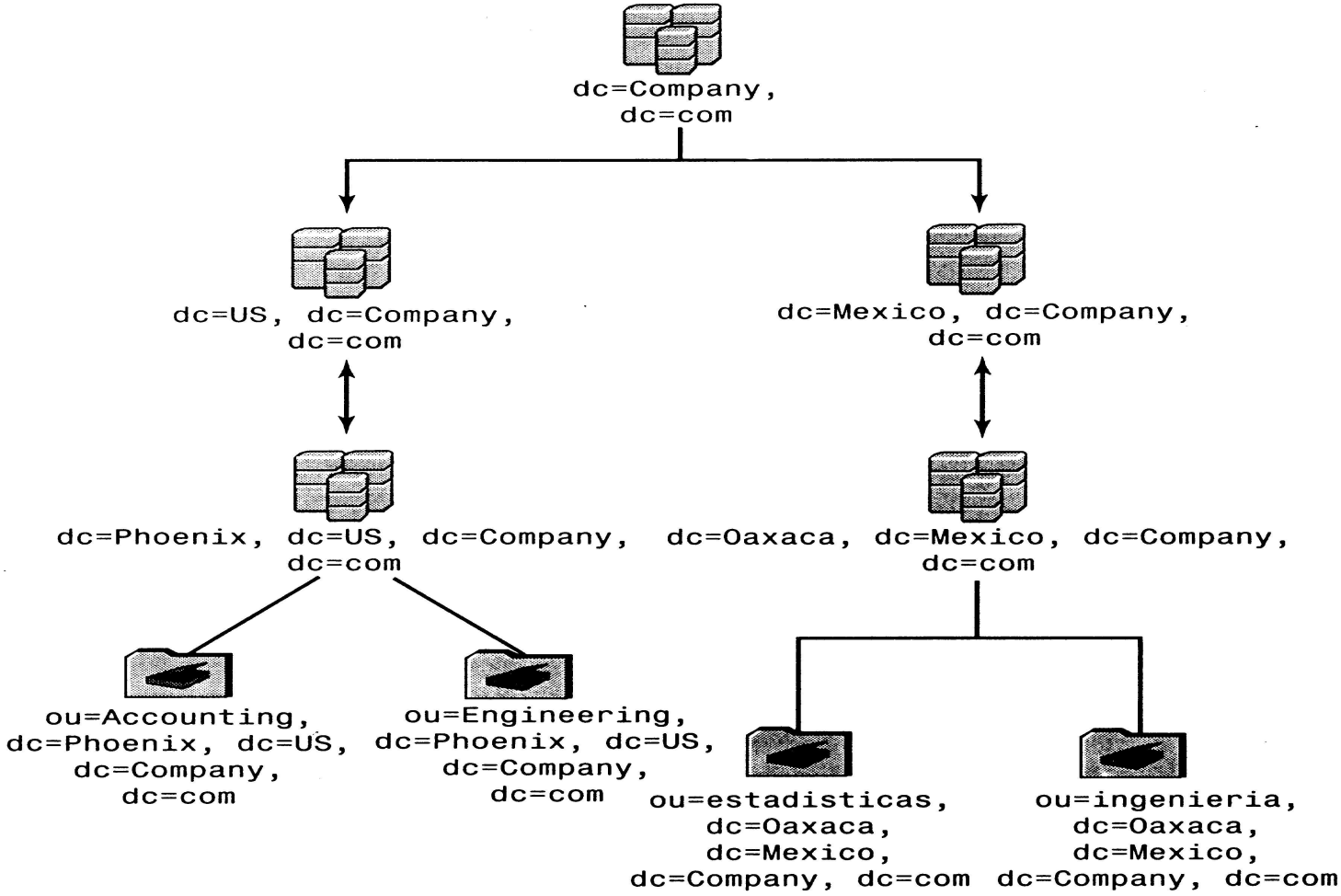
- A protocol for managing and querying information in a directory service over a TCP connection (port 389)
- Available on a range of platforms (Windows, Linux)
- Implemented with object-oriented database
- Information stored in *directory information tree* (DIT)
- Each object in the DIT called a *directory service entry* (DSE)
- Each DSE in a DIT has a unique name
  - *distinguished name* (DN) - entire name from root of directory
  - *relative distinguished name* (RDN) - no part or partial path to root

LDAP Naming Conventions

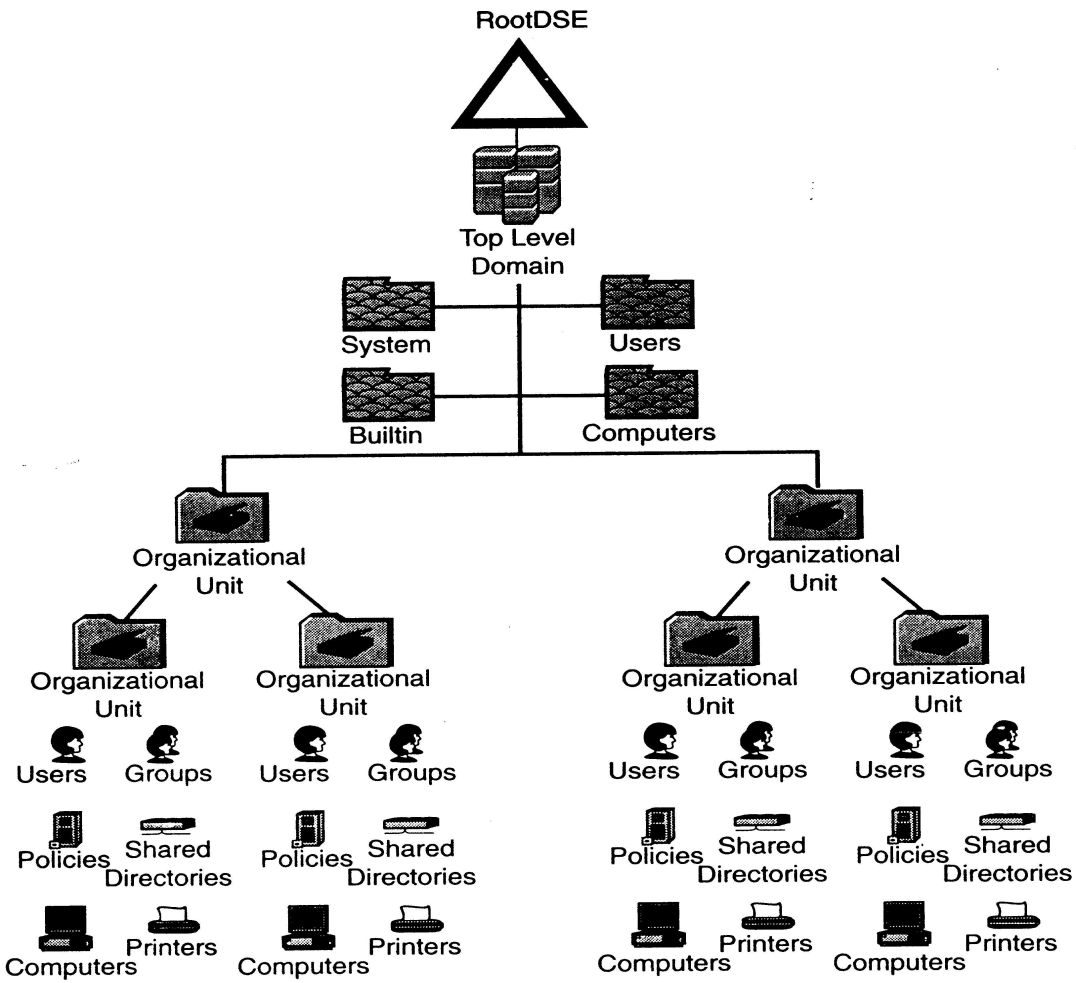
- Simple name of an object is its *common name* (CN)
- Domain objects are named with *domain component* designators (DC)
- Organizational unit objects are named with OU designators
  - Organize the database to reflect company organizations
- Example (names are not case sensitive):
  - Our domain is dc=infosys, dc=bcit, dc=ca
  - A user account for 00185586 would be

    cn=00185586,cn=users,dc=infosys,dc=bcit, dc=ca

Distinguished Name Format



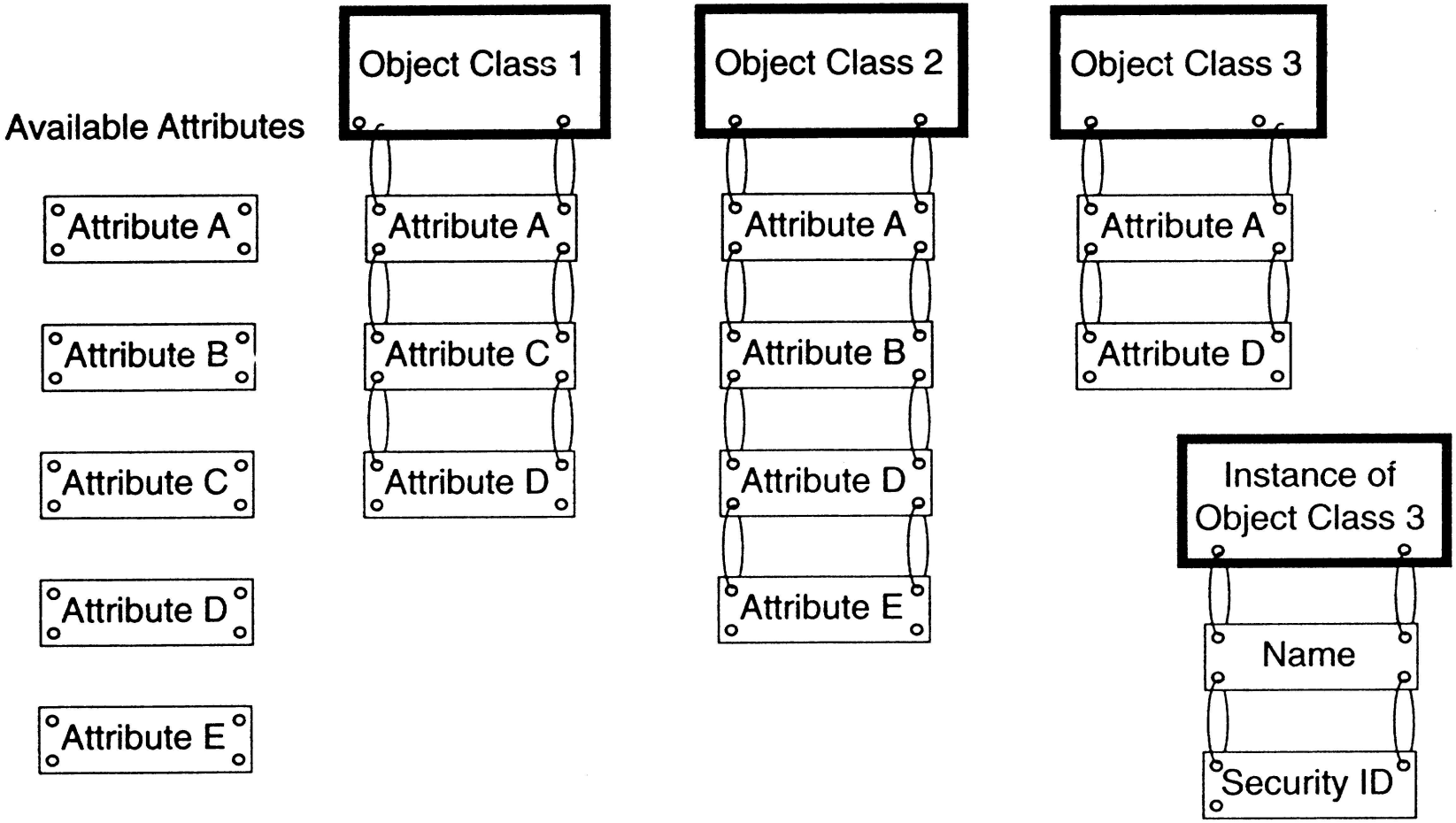
Example Directory Container Structure for Domain



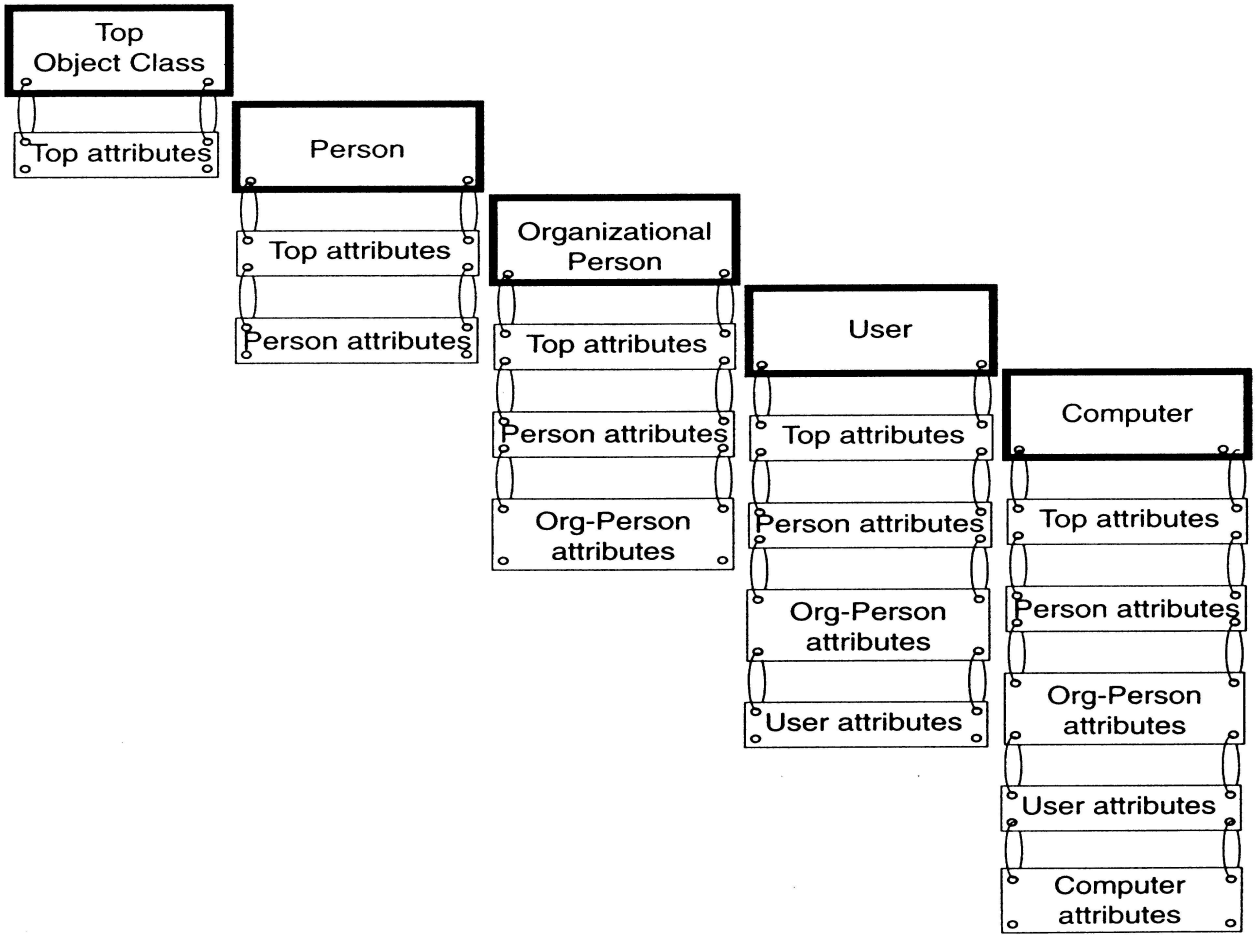
Entries

- *Entry* (node in LDAP hierarchy) basic unit of information
- Entry has one or more *attributes*
  - Attribute has name and one or more values
  - Attribute names defined in schema files stored in /etc/openldap/schema directory
    - See, for example, core.schema
- Entities inherit attributes from their parent types
  - Schema defines which attributes are mandatory
- Entries can be stored in LDAP Data Interchange Format files (LDIF)

Classes and Attributes In a Directory Service



Object-Oriented Database - Computer Object



Setting Up LDAP Server

- Install openldap-clients and openldap-servers
- Start the ldap service
  - This starts the slapd daemon (!)
- For network use, slapd uses port 389 for unencrypted, or port 636 for encrypted queries
  - Open appropriate port through firewall
- Setting up the hierarchy takes some thought about data being stored and how to organize
  - Example in text is good to play with

Configuring LDAP Server /etc/openldap/slapd.conf

- Suffix line defines DSE
  - `suffix  "dc=sobell, dc=com"`
- rootdn is the LDAP administrator
  - `rootdn "cn=ldapadmin, dc=sobell, dc=com"`
- rootpw sets password for administrator
  - `rootpw  porcupine`
- Can use `slappasswd` to encrypt a password for you to paste into rootpw line

LDAP Utilities

- ldapsearch – search for entry
- ldapadd – add entry
- ldapmodify – modify entry (can also add)
- /usr/sbin/slapcat – display all entries in LDIF format
  - Must be run as root
- Many email and contact management clients can retrieve data from LDAP database
  - kaddressbook, browsers ldap://, gq LDAP client

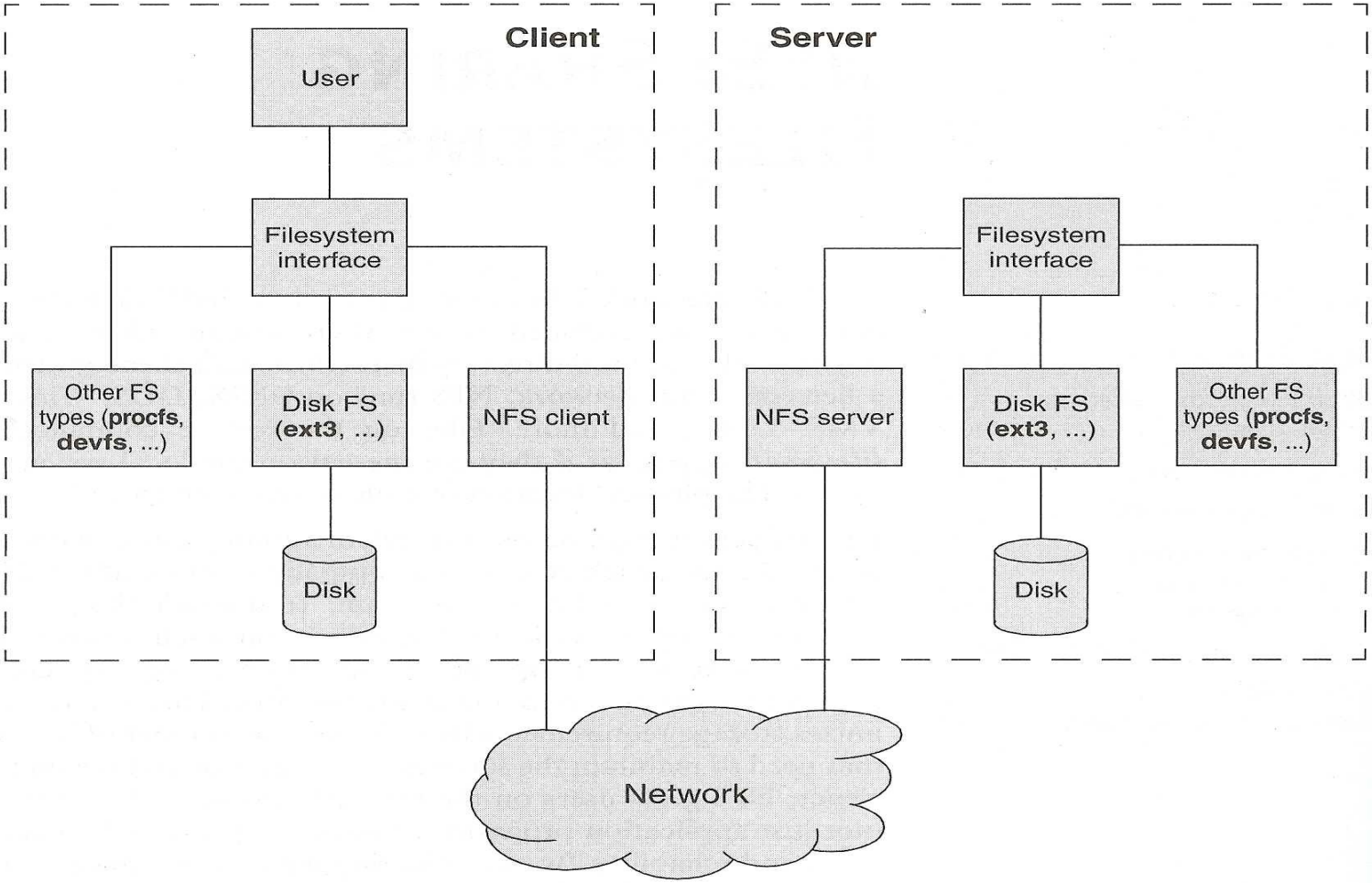
NIS/LDAP Summary

- NIS (Network Information Service) maintains common administrative files in central database and provides to clients
  - Files converted to map files to be distributed
  - Can display contents with ypcat and ypmatch
  - yppasswd utility replaces passwd for changing user passwords
- LDAP (Light-weight Directory Access Protocol) maintains a search and read optimized database

Chapter 22: NFS: Sharing Filesystems

- Network Filesystem protocol, UNIX standard
- Files on remote server appear as if present on client
- Allows administrator to centralize copies of application programs
- Can set up diskless systems with all software/data centrally located
- Can set up dataless systems where user data centrally stored

Data Flow in NFS Setup



Setting up NSF Client

- Install nsf-utils and system-configfs
  - No daemons need to be started
- Use mkdir to create mountpoints
- Use mount to mount directory
  - mount -o ro, nosuid castor:/opt /opt
  - See text for additional options
- Create entries in /etc/fstab if use frequently
- Can umount remote directory when done with it

Setting Up an NFS Server

- Install nfs-utils, system-config-nfs
- Start nfs service when system enters multiuser mode
  - /sbin/chkconfig nfs on
- Open TCP port 111 for portmap, TCP ports 1013 and 1016 for mountd and TCP port 2049 for NFS
  - If still does not work, try `rpcinfo -p` to see what ports are being used

Using system-config-nfs

- This command give you a GUI-based wizard for setting up exports file
  - Discussed exports file in earlier NFS introduction
- Allows specifying which directory hierarchies are shared and how
  - Add share by clicking Add
  - Modify share by clicking Properties
    - Opens Edit NFS Share window
  - Both add and edit windows have same sections

Settings for a Share

- Basic tab
  - Path to shared directory, hosts that can mount share (names or IPs), ro or rw permission
- General Options tab
  - Allow connections from insecure ports
  - Allow insecure file locking
  - Disable subtree checking
  - Sync write operations on request
  - Force sync or write operations immediately
  - Hide filesystems beneath (mounted in exported directory)
  - Export only if mounted

Settings for a Share

- User Access tab
  - Treat remote root user as local root (don't do)
  - Treat all client users as anonymous users
    - This is probably too drastic
    - Be sure user/group IDs are same on client and server (use NIS or LDAP)
  - Local user ID for anonymous users
  - Local group ID for anonymous users
- All changes are made to /etc/exports file
  - Can edit by hand as of olden times

Automatic Mounting: automount

- Allows mounting directories when referenced
- Install autofs package and start autofs when system enters multiuser mode
  - autofs is a directory hierarchy which stays unmounted until needed
    - Unmounted after being inactive (default 5 minutes)
- Main control file is `/etc/auto.master`
  - Three columns:
    - Parent directory of autofs mountpoint
    - name of map file with associated information
    - optional column of mount options
  - E.g. `/mount /etc/auto.misc –timeout 60`

Format of Automount Map File

- Map file can have any name
- Three columns
  - name of autofs mountpoint under parent dir
  - options
  - server and directory hierarchy to be mounted
- E.g. `fred -fstype=ext3 castor:/home/fred`
- Running `/sbin/service autofs status` shows configured and active autofs mount points

NFS Summary

- NFS (Network Filesystem) allows server to share local directory hierarchies
- On server the /etc/exports file lists the exported directory hierarchies
- From client can mount exported NFS directory hierarchy
  - Or put entry in /etc/fstab to automatically mount when system boots
- Can set up as auto mounted system which is auto dismounted when not used

Chapter 23: Samba: Integrating Linux and Windows

- Samba allows UNIX-like operating systems to work with OS/2 and Windows as client or server
- Can use Samba to run a Windows Print/File server using a Linux server
  - Perhaps more efficient than using Windows server

Installing Samba

- Install samba, samba-client, samba-common, system-config-samba, samba-swat
- Configure `smb` daemon to start in multiuser mode
- To use swat, modify /etc/xinetd.d/swat to enable and restart xinetd
- Samba uses UDP ports 137, 138 and TCP ports 139 and 445
  - Firewall Configuration wizard has Samba checkbox

Samba Utilities and Daemons

- net: same as DOS net command
- nmbd: NetBIOS nameserver program
- nmblookup: makes NetBIOS name queries
- smbclient: displays shares on a Samba server
- smbd: Samba daemon provides file/print services
- smbpasswd: changes Windows NT password hashes
- smbstatus: displays information about smbd connections
- smbtree: displays hierarchical diagram of shares
- swat: Samba Web Administration Tool
- testparm: checks syntax of smb.conf file

Samba Users, User Maps and Passwords

- Windows user provides Windows username and Samba password
  - Windows will send its username/password to authenticate, then prompt user if that fails
  - Can also use LDAP or PAM to authenticate
- Windows username must be same as Linux username or map to Linux username
  - Maps kept in /etc/samba/smbusers
  - Format: *`linux-user: samba-user-list`*
- Samba passwords kept in /etc/samba/smbpasswd
  - `smbpasswd -a` *`username`*

Configuring Samba Server With system-config-samba

- Easiest way to get started
  - Configures server
    - Under Preferences>Server Settings
    - Define workgroup, set authentication to User and set Encrypt Passwords to Yes
  - Configures users
    - Under Preferences>Samba Users
    - Add/Edit/Delete users
    - As root, can select Linux user; set Windows user/password
  - Sets up shares: click on Add Share on toolbar

Configuring Samba Server With swat

- swat is browser-based graphical editor for /etc/samba/smb.conf file
  - Overwrites, so make copy first
  - Provides help, default values and forms to change values
- Started from xinetd as mentioned earlier
  - If want to access remotely open TCP port 901
    - Bad idea if insecure network or over Internet due to password sniffing
- Go to http://localhost:901, login as root

swat Home Page Buttons

- HOME: displays links to local documentation
- GLOBALS: Edits global variables in smb.conf
- SHARES: Edits share information in smb.conf
- PRINTERS: Edits printer information in smb.conf
- WIZARD: Rewrites smb.conf removing comments and default values
- STATUS: shows active connections, controls smbd and nmbd
- VIEW: Displays configuration parameters in smb.conf
- PASSWORD: Manages passwords

swat Help and Defaults

- If you are going to use Samba, swat is a good place to start
- Home page points to detailed local documentation
- Each variable/parameter has help link that opens a window explaining the parameter
  - Also Set Default button to reset the variable/parameter to default value
- Click Commit Changes before leaving a page

Setting Global Section with swat

- Click GLOBALS button on home page
- Set **`workgroup`** to be name of workgroup
- Set **`hosts allow`** to names or IP addresses of machines than can access shares
  - include localhost
  - separate list with spaces or commas
- Set **`hosts deny`** to ALL
- Click **`Commit Changes`**

Defining Share with swat

- Click SHARES button
- Enter name of share in box next to **`Create Share`** button and click **`Create Share`** button
- Information about the share is displayed
- Enter **`path`** (absolute path to shared directory on local Linux) and optional **`comment`**
- Click **`Commit Changes`**
- Clicking on **`Advanced`** button displays many other parameters you can set

Manually Configuring Samba Server

- Edit the /etc/samba/smb.conf file directly
- smb.conf has extensive comments and commented examples
  - Comments start with # or ;
- smb.conf file divided into sections named in [ ]
  - [globals] defines global parameters
  - [printers] defines printers
  - [homes] defines shares in homes directory
  - [*share name*] defines a share (one section each)

Parameters in smb.conf

- Lots of them, see text
- **`workgroup`**: defines workgroup server belongs to
- **`encrypt passwords`**: used to require encrypted passwords from users
- **`hosts allow`** and **`hosts deny`**: used to control hosts that can access shares
- [homes] share makes each user's home directory shared to those users

Accessing Shares from Windows

- Like is normally done in windows
- Open My Computer or Explorer
- In Address text box enter *hostName*
  - Windows displays the shares from *hostName*
  - If share is not browsable, enter \\hostName\shareName
- Can also map share to drive letter using Map Network Drive (Tools menu in Windows XP)

Accessing Shares from Linux

- `smbtree` displays hierarchical diagram of available shares
- `smbclient` connects to Windows shares
  - `smbclient //hostName/shareName`
  - After connecting can use ftp-like commands
- Gnome and KDE have more friendly alternatives
  - From Konqueror or Nautilus enter smb:/ in the location bar to browse Windows shares
  - Other native Gnome or KDE applications as well
- Can use mount with `-t cifs` to mount window share under Linux hierarchy

Troubleshooting Samba

- Restart smbd and nmbd daemons and check for successful start
- Run `testparm` to see if smb.conf is syntactically correct
- Use ping to make sure machines are connected over the network (both ways)
- From Windows command prompt run **`net view \\hostName`** to display list of shares available from server
- Try to map drive from Windows command prompt: **`net use x: \\`*`hostName`*`\`*`shareName`*

Troubleshooting Samba

- From server, query nmbd server using __SAMBA__ for server's NetBIOS name
  - `nmblookup -d 2 -B` *`hostName`* `__SAMBA__`
- From server, query nmbd server for client's NetBIOS name
  - `nmblookup -B` *`clientName`* `\*`
- From server, generate a list of shares from smbclient
  - `smbclient -L serverName`
- Query for master browser from server
  - `nmblookup -M` *`WORKGROUP`*

Samba Summary

- Allows Windows to share files and printers using SMB protocol
- User names correspond to Linux user names
  - Directly or mapped by smbusers
- Separate smbpasswd file used for passwords
- Main configuration in smb.conf
  - Simple configuration with system-config-samba
  - More complex configuration with swat
- Windows client accesses shares as usual